

Likhith Kumar Mankala

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SUMMARY

AI/ML Engineer and Full-Stack Developer with hands-on experience building LLM-powered applications, RAG pipelines, agentic AI systems, and production-grade web platforms. Proven track record deploying deep learning models (CNN/BiLSTM/Transformers), LLM integrations, and scalable REST APIs serving thousands of users. Research co-author (Elsevier preprint) on sign language recognition achieving 94.7% accuracy. Mentor to 250+ students across AI bootcamps and workshops. Passionate about bridging research and real-world AI deployment.

TECHNICAL SKILLS

LLMs & AI	OpenAI API, Gemini, LangChain, RAG Pipelines, Llama 3.1 (Ollama), Agentic AI Frameworks, Prompt Engineering, Fine-tuning, Vector DBs
ML / DL	PyTorch, TensorFlow, Keras, CNN, BiLSTM, Transformers, Multi-Head Attention, YOLOv8, MediaPipe, DeepFace, Scikit-learn, OpenCV
Languages	Python, TypeScript, JavaScript, Java, SQL, C, Dart (Flutter)
Web / Backend	Next.js, React, NestJS, FastAPI, Flask, Django, REST APIs, WebSockets, SSR, React Native, Flutter
Databases	PostgreSQL (Neon), MySQL, MongoDB, Drizzle ORM, Supabase
Tools & DevOps	Docker, Jenkins, Git, GitHub Actions, Vercel, Qiskit, Figma, Jupyter, Kaggle, Agile/Scrum, System Design

PROFESSIONAL EXPERIENCE

AI Full-Stack Intern — Qubitedge Global Services	Apr 2026 – May 2026
- Built and deployed multiple AI/LLM-powered full-stack applications including a RAG-based document Q&A; system, an agentic task automation tool, and an LLM chatbot integrated into the company LMS portal. - Developed production REST APIs with FastAPI and Next.js SSR front-ends; integrated OpenAI and Llama 3.1 models with LangChain for intelligent response generation and retrieval pipelines. - Led AI bootcamp sessions and mentored 250+ students across Applied AI & Data Science programs; created hands-on project modules covering LLMs, computer vision, and full-stack AI deployment.	
AI/ML Intern — UptoSkills	Feb 2026 – May 2026
- Built and deployed classification and prediction ML models using Python, TensorFlow, and Scikit-learn; integrated model outputs into Flask REST API endpoints for real-time inference. - Developed data pipelines for preprocessing and feature engineering, improving model accuracy across multiple projects; contributed to NLP and computer vision modules.	
AI & Data Analytics Intern — Edunet Foundation – Skill4Future	Jun 2025 – Aug 2025
- Built AI e-waste classification system (TensorFlow/PyTorch) achieving 92% accuracy; deployed Flask web interface for live inference. - Developed EV demand forecasting model on 50K+ sales records with 87% accuracy; reduced ML deployment time by 40%; optimized deep learning inference 35% via quantization and pruning.	
Software Development Intern — NTS Nihon Global	Jul 2025 – Aug 2025
Built 12+ PostgreSQL-backed REST APIs supporting 10K+ daily users; improved engagement 28% via UI/UX enhancements; reduced load time 45% by debugging and optimizing legacy codebase.	
Python & Web Developer Intern — YoungBot Academy	Oct 2024 – Dec 2024
Automated data pipelines (Pandas/Scrapy) processing 100K+ records; designed React UI components lifting engagement by 42%; reduced page load times 55% via code splitting and lazy loading.	

KEY PROJECTS

SignBridge – Real-Time Continuous Sign Language Recognition (Elsevier Preprint) | MediaPipe, CNN-BiLSTM, Attention, DeepFace, Llama 3.1, Flask, React

- End-to-end SLR framework: MediaPipe Holistic (1,662-dim landmarks) → CNN spatial encoder → BiLSTM-Attention temporal model → CTC decoding; achieved **94.7% word accuracy** and 12.3% WER on ISL-50 dataset (5,000 clips, 50 classes).
- Integrated DeepFace emotion recognition + local Llama 3.1 LLM for emotionally-nuanced natural language sentence synthesis with offline pyttsx3 TTS output; real-time inference at **28ms/frame** on GPU.

ClimateChain – Quantum-Secured Climate Data Platform | Next.js, Qiskit, Python, TensorFlow | aqvh-2025.vercel.app

- Quantum-secured blockchain ledger processing 10K+ daily climate transactions; AI drought prediction model achieving **89% accuracy** for 1,000+ researchers. Top 5 Finalist – AP Quantum Valley Hackathon.

University Medical Assistant (LLM-Powered RAG Portal) | Python, Flask, LangChain, PostgreSQL, OpenAI API

- HIPAA-compliant medical portal with fine-tuned LLM + RAG pipeline achieving **85% symptom classification accuracy** for 5,000+ students; context-aware responses from medical knowledge base.

QubitEdge LMS Portal | Next.js Full-Stack, PostgreSQL, TailwindCSS | lms.qubitedge.in

- Full-stack LMS with Next.js App Router SSR, course management, student enrollment, progress tracking, and role-based dashboards for admins, instructors, and learners; subdomain deployed via Vercel.

DSNLU University Website & ITYukta 2K26 Tech Fest | Next.js, NestJS, TypeScript, PostgreSQL | dsnlu-frontend.vercel.app | ityukta2k26.jntugvceev.in

- Lead designer & developer of two official institutional platforms; built scalable NestJS APIs with RBAC, PostgreSQL CMS, event listings, registration flows, and responsive UI/UX.

Flutter Multiplayer Game | Flutter, Dart, Firebase, WebSockets

- Real-time multiplayer mobile game with Firebase auth, WebSocket game-state sync, matchmaking logic, and leaderboard; deployed on Android & iOS.

EDUCATION

B.Tech in Information Technology — JNTUGV Vizianagaram | CGPA: 8.30 | Oct 2023 – Apr 2027

ACHIEVEMENTS & LEADERSHIP

- Top 5 Finalist** – AP Quantum Valley Internal Hackathon (20+ teams) | Participated in SIH 2025, Miracle Citizen Hack 2025, GenAI 24-Hr Hackathon
- Tech Lead – Yukta Dev Squad**: Led 8-member team delivering 5+ production apps; mentored 6 junior developers (↑35% team productivity)
- Founder – Qubitedge Global Services**: Built and operated AI bootcamp programs; mentored **250+ students** in AI, LLMs, and full-stack development
- Scaler Certified: Data Science, JavaScript, React.js, ML Engineering | LeetCode 100-Day & 50-Day Badges | Runner-up WWF State Quiz 2018 (1,000+ participants)